

RAQIB — AI targeting of customs controls

Team RAQIB · Hackathon « IA & Finances Publiques » 2026 · Prototype designed for Tunisian Customs · Advisory tool

+22% more frauds caught with the same 470 inspections (317 vs 259)	5.9× public-safety threats caught (41 vs 7 for the best rule)	−19% inspections needed to catch the rule's frauds (379 vs 470)	12 / 13 test weeks in which the AI beats the rule
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1. The challenge

Customs can physically inspect only a small share of import declarations. Selection today relies on fixed rules (for example the product's past fraud rate) and on officers' experience, so many inspections find nothing while duty fraud and dangerous goods are released. **RAQIB decides every day, within a fixed inspection capacity**, which declarations to inspect (**RED**), to check on documents (**YELLOW**) and to release (**GREEN**), for two objectives: **revenue** (duty fraud) and **public safety** (critical violations). It shows, side by side, what the best current rule would have chosen, and leaves the final decision to the officer.

2. Data

Public **Customs Import Declaration Datasets** (Institute for Basic Science & Korea Customs Service, MIT licence): 54,000 synthetic declarations generated (CTGAN) from real inspected declarations, 22 attributes including the fraud and critical-fraud outcomes; 14,482 importers, 2,489 products, 119 origins. **Train**: 45,519 declarations (Jan 2020 → Mar 2021). **Test**: 8,481 declarations (Apr → Jun 2021), never used for training. Product names: datasets/harmonized-system (ODC-PDDL). Public Tunisian statistics (UN Comtrade 2024) are shown in the app as context only and are **never a model input**; no access to any administration system.

3. Technical approach and models

- Risk-history features.** For 8 keys (product HS6 / HS4 / HS2, importer, declarant, seller, origin, office): the smoothed past fraud rate and volume, computed *out-of-fold* (5 folds) so no row ever sees its own label; plus tax rate, net mass, value and unit value.
- Two supervised models** learned from past inspection outcomes: duty fraud (primary: glass-box **EBM**) and critical fraud (primary: **LightGBM**). Each has a twin (EBM / LightGBM); when the twins disagree, the case is flagged for a human.
- Calibration** (isotonic regression): scores become probabilities an officer can read.
- Daily allocation**: capacity = 5% of the day's declarations (adjustable). Public-safety alerts (top 1% critical risk) take slots first, then the highest fraud probabilities; optional 10% random **exploration** keeps the system learning; the next 10% go to YELLOW, the rest to GREEN.
- Explanations**: exact additive contributions (EBM terms or TreeSHAP) grouped into the 4 strongest plain-language reasons with the real historical numbers, and an exact waterfall.
- Human in the loop**: the officer decides; each decision is appended to a hash-chained, tamper-evident journal.

Stack: Python (pandas, scikit-learn, LightGBM, InterpretML EBM), FastAPI, React. The full pipeline trains in about 15 s on a laptop CPU; a declaration is scored in about 0.1 s; the demo runs fully offline. An optional local LLM only rephrases computed facts; it never scores, ranks or decides.

4. Results (test period, never used for training)

Ranking quality — share of fraud among the top k% ranked declarations (precision) and share of critical cases found (recall):

Method	Fraud AUC	Fraud prec. @1%	@5%	@10%	Critical AUC	Crit. recall @1%	@5%	@10%
RAQIB AI	0.771	84.7%	71.8%	61.4%	0.952	52.1%	78.1%	84.9%
Rule: product (HS6) history	0.728	57.6%	53.4%	48.6%	0.924	26.0%	54.8%	75.3%
Rule: importer history	0.513	20.0%	18.8%	19.7%	0.516	0.0%	1.4%	6.8%
Random	0.497	24.7%	21.4%	20.4%	0.548	0.0%	6.8%	13.7%

- Duty-fraud precision @5%: 71.8% vs 53.4% for the rule, **+17.8 pts** (95% bootstrap CI +12.5 to +22.5 pts).
- Public-safety recall @5%: 78.1% vs 54.8%, **+22.5 pts** (95% CI +13.3 to +32.3 pts; only 73 critical cases, ±5 pts).
- Stable (AI beats the rule in 12 of 13 weeks) and calibrated (riskiest tenth: 61.7% predicted, 61.4% observed).

Daily replay — 91 real test days, identical capacity (5% of each day's declarations) for every policy:

Policy	Inspections	Frauds caught	Threats caught	Hit rate	Distinct products
RAQIB AI	470	317	41	67.4%	261
RAQIB AI + 10% exploration	470	294	41	62.6%	280
Current rule (product history)	470	259	7	55.1%	163
Random	470	103	3	21.9%	309

- **Same workload, more results:** +58 frauds (+22%) and 5.9× the public-safety threats versus the best rule.
- **Same frauds, fewer inspections:** the rule needs 470 inspections for 259 frauds; RAQIB catches 263 with 379 (**19% fewer**; pooled ranking: 302 vs 425, 29% fewer).
- **Glass box, no accuracy lost:** EBM duty-fraud AUC 0.771 / precision @5% 71.8% vs 0.770 / 71.1% for the black-box LightGBM.
- **Uncertainty goes to a human:** 371 test declarations (4.4%) where the twin models disagree have a 44% fraud rate (average 22%); 136 of them move from GREEN to a document check. 82% of declarations are released GREEN.

5. Limits identified

- Synthetic data of Korean origin; only inspected declarations were synthesised, so the fraud rate (21.6%) is far above reality: absolute precision is optimistic. **Our claim is the gain over the rule on the same data.**
- Only 73 critical cases in the test period (±5 pts on recall @5%).
- Value and mass carry signal here (without them, LightGBM precision @5% falls from 71.1% to 63.3%), but 78% of test declarations have exactly their product's usual unit value: to re-validate on real data.
- 10% of test declarations come from never-seen importers, treated as average risk.
- Learning only from inspected declarations creates selection bias; exploration and a control group correct it.
- Alert and lane thresholds were set on the test period; in production they must come from previous weeks.
- Tested and rejected: anomaly detection (Isolation Forest, fraud AUC 0.48) and 2-hop network features (no gain).

6. Recommendations for production in Tunisia

1. **Retrain inside the administration** on Tunisian inspection outcomes (same recipe; minutes on a standard CPU); no data leaves.
2. **Shadow mode (4–8 weeks):** RAQIB scores in parallel with current rules, without affecting any decision.
3. **Controlled pilot:** some offices or days use RAQIB lanes, others keep the rules; measure hit rate, duties recovered, threats found and clearance time.
4. **Keep 5–10% exploration** and a random control sample so the model stays unbiased.
5. **Weekly monitoring:** drift, calibration, share of new operators, selection rates by office and transport mode; retrain on drift.
6. **Governance:** advisory score only, officer decides, every decision journaled; compliance with Organic Law 2004-63 on personal data and review by the INPDP.

Third-party components: all permissive licences (full list in THIRD_PARTY.md). Code and demo: github.com/yassinebennacef/RAQIB. Claude Code (Anthropic) was used as a coding assistant; scoring uses only the models above.